CANARA COLLEGE LIBRARY MANGALURU-3

Reg. No. | | | | | | | | | | | | |

BCAAIMS 202

II Semester B.C.A. (AI and ML) Degree Examination, June/July 2026
(SEP) (2024 – 25 Batch Onwards)
DATABASE MANAGEMENT SYSTEMS

Time : 3 Hours Max. Marks : 80

***Note** : Answer **any ten** questions from Part – **A** and **one full** question from **each** Unit in Part – **B**.*

PART – A **(10×2=20)**

1. a) What is database ? Define DBMS.
   b) Define Metadata. Give an example.
   c) List any two responsibilities of DBA.
   d) Define Single Valued Attribute. Give an example.
   e) Give any two basic operations that can change the state of relations in the database.
   f) Define Primary key. Give an example.
   g) What is dual table ? Give an example.
   h) What is the use of ORDER BY clause ? Write the syntax.
   i) List any four cursor attributes.
   j) How can you change the name of the table ? Give an example.
   k) What is the purpose of ROLLBACK command ?
   l) List any four types of join used in SQL.

PART – B

Unit – I

2. a) Explain the advantages of DBMS.
   b) Explain the three schema architecture of a database with a neat diagram.
   c) Write a note on Database system utilities. **(5+5+5)**

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3. a) Explain strong and weak entity types in DBMS with example.
   b) What is data Independence ? Explain their types.
   c) What are the characteristics of DBMS ? Explain. **(5+5+5)**

Unit – II

4. a) Explain 1NF with example.
   b) Explain the term Domain, Tuple and Relation schema with example.
   c) Write a note on key constraints. **(5+5+5)**

5. a) Explain the characteristics of Relation.
   b) Discuss Entity integrity and referential integrity constraints.
   c) Explain functional dependencies with example. **(5+5+5)**

Unit – III

6. a) Explain CHECK constraint with syntax and example.
   b) Explain any five aggregate functions with syntax and examples.
   c) Create the table : STUDENT

Column Name	Data Type	Size	Attributes
REG_NO	Varchar2	5	Primary key First letter must start with S
NAME	Varchar2	20	Not Null
CLASS	Varchar2	4	BCA/BBA/BCOM
BIRTH_DATE	Date	8	
SEX	Character	1	M/F
HEIGHT	Number	6, 2	NOT NULL
WEIGHT	Number	6, 2	NOT NULL

Write the following queries :

i) List the name of the boys with weight more than 55 kgs.
ii) List the details of students in alphabetical order of name.
iii) List the maximum, minimum and average weight of students in each class. **(5+5+5)**

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7. a) Explain any five clauses used in SELECT statement with example.

   b) Explain any five Scalar functions with syntax and example.

   c) Create the table : CLIENT_MASTER

Column Name	Data Type	Size	Attributes
CLIENT_NO	Varchar2	6	Primary key First letter must start with C
NAME	Varchar2	20	
ADDRESS1	Varchar2	30	
ADDRESS2	Varchar2	30	
CITY	Varchar2	15	
PINCODE	NUMBER	8	NOT NULL
STATE	Varchar2	15	NOT NULL
BALDUE	NUMBER	10, 2	NOT NULL

Consider the table CLIENT_MASTER given, write the following queries :

i) Find out the name of the clients with CLIENT_NO='C00001'.

ii) Change the city of CLIENT_NO C00005 to 'Bangalore'.

iii) Add a column called 'TELEPHONE' of data type NUMBER and size = 10 to the table CLIENT_MASTER. **(5+5+5)**

Unit – IV

8. a) What is trigger ? Explain different types of triggers.

   b) Explain various if...end if structures with syntax and example.

   c) Explain the structure of PL/SQL program with example. **(5+5+5)**

9. a) What is cursor ? Explain different types of cursors.

   b) With syntax and example explain how a package is created and executed in PL/SQL.

   c) List any five system exceptions of PL/SQL and explain. **(5+5+5)**

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